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SYSTEM AND METHOD FOR DELIVERING BEAMS TO PREDETERMINED LOCATIONS OF WORK SITES

Abstract

A system for delivering beams to predetermined locations of a work site includes a sorting assembly, a work machine, and a control system. The sorting assembly receives the beams in loose batches and arranges them in a predefined arrangement. The work machine includes an implement having haul elements. The control system receives an instruction to transfer the beams to the predetermined locations and issue a command to move the work machine for operative engagement with the sorting assembly. Further, the control system generates a retrieval command to alter the implement and switch the haul elements to a first condition to retrieve the beams from the sorting assembly. Also, the control system provides a notification to haul the beams to the predetermined locations. The control system generates release commands at the predetermined locations to release at least one beam correspondingly at the predetermined locations in a preset orientation.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to delivering beams to predetermined locations of a work site. More particularly, the present disclosure relates to a system and method for delivering beams to the predetermined locations of the work site, at least in part, by using autonomous or semi-autonomous machines.

BACKGROUND

[0002] A work site, such as a solar farm, often requires large quantities of materials or components, such as torsion bars, beams, and/or the like (e.g., to support solar panels) to be delivered to various designated or predetermined locations of the work site. While devices, such as the solar panels, may be delivered in crates, components, such as, beams may be delivered in loose batches to the predetermined locations by one or more supply machines. Once a supply machine reaches a designated or a predetermined location, workers are typically required to manually unload the beams from the supply machine and place them at the predetermined locations in a desired orientation. Such conventional methods for delivering such components at those locations are time-consuming and labor-intensive.

SUMMARY OF THE INVENTION

[0003] In one aspect, the disclosure is directed towards a system for delivering one or more beams to one or more predetermined locations of a work site. The system includes a sorting assembly, a work machine, and a control system. The sorting assembly is configured to receive the beams in loose batches and arrange the beams in a predefined arrangement. The work machine includes an implement having one or more haul elements mounted thereto. The control system is configured to receive an instruction to transfer the beams to the predetermined locations and issue a command to move the work machine for operative engagement with the sorting assembly in response to the instruction. Further, the control system is configured to generate a retrieval command to alter the implement and switch the haul elements to a first condition to retrieve the beams from the sorting assembly in the predefined arrangement. Also, the control system is configured to provide a notification to haul the beams in the predefined arrangement to the predetermined locations. The control system generates release commands correspondingly at the predetermined locations such that at least one haul element switches from the first condition to a second condition to release at least one beam correspondingly at the predetermined locations in a preset orientation.

[0004] In another aspect, the disclosure relates to a method for delivering one or more beams to one or more predetermined locations of a work site. The method includes receiving an instruction to transfer the beams to the predetermined locations and issuing a command to move a work machine for operative engagement with a sorting assembly in response to the instruction. The sorting assembly receives the beams in loose batches and arranges the beams in a predefined arrangement. The work machine includes an implement having haul elements mounted thereto. The method further includes generating a retrieval command to alter the implement and switch the haul elements to a first condition to retrieve the beams from the sorting assembly in the predefined arrangement. Further, the method includes providing a notification to haul the beams in the predefined arrangement to the predetermined locations. The method additionally includes generating release commands correspondingly at the predetermined locations such that at least one

haul element switches from the first condition to a second condition to release at least one beam correspondingly at the predetermined locations in a preset orientation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a plan view of an example work site and a system for delivering beams to predetermined locations of the work site, in accordance with an embodiment of the present disclosure;

[0006] FIG. **2** is a perspective view of an example work machine having an implement, in accordance with an embodiment of the present disclosure;

[0007] FIG. **3** is another perspective view of the work machine having one or more haul elements mounted onto the implement, in accordance with an embodiment of the present disclosure;

[0008] FIG. **4** is yet another perspective view of the work machine illustrating beams grasped and retained by the haul elements, in accordance with an embodiment of the present disclosure;

[0009] FIG. **5** is a front view of a single haul element, in accordance with an embodiment of the present disclosure;

[0010] FIG. **6** is a top view of the single haul element of FIG. **5**, in accordance with an embodiment of the present disclosure;

[0011] FIG. **7** is a side view of the single haul element of FIG. **5**, in accordance with an embodiment of the present disclosure;

[0012] FIG. **8** is a top view of a sorting assembly, in accordance with an embodiment of the present disclosure;

[0013] FIG. **9** is a side view of the sorting assembly of FIG. **8**, in accordance with an embodiment of the present disclosure;

[0014] FIG. **10** is a schematic view of an example control system in connection with systems and sub-systems of the work machine of FIGS. **2** through **4** and the sorting assembly of FIGS. **8** and **9**, in accordance with an embodiment of the present disclosure; and

[0015] FIGS. **11A-C** are block diagrams of example systems for delivering beams to a predetermined location of the work site;

[0016] FIG. **12** is a flowchart illustrating an example method for delivering the beams to one or more predetermined locations of the work site, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0017] Reference will now be made in detail to specific embodiments or features, examples of which are illustrated in the accompanying drawings. Generally, corresponding reference numbers may be used throughout the drawings to refer to the same or corresponding parts, e.g., **1**, **1'**, **1''**, **101** and **201** could refer to one or more comparable components used in the same and/or different depicted embodiments.

[0018] Referring to FIG. **1**, an example work site (see work site **100**) is shown. Although not limited, the work site **100** may be a construction site. In one example, the work site **100** may be a construction site for the construction of a solar farm **104**, and one or more segments of a construction process for building the solar farm **104** will be discussed, by example, in the present disclosure. Without limitation, however, various other applications of the work site **100** may be contemplated by someone skilled in the art, based on the present disclosure. The work site **100** may include a construction area **108** and a storage area **112**.

[0019] With continued reference to FIG. **1**., the storage area **112** (also referred to as a storage facility or a staging area) of the work site **100** may include or correspond to a region of the work site **100** where materials **116** applicable for the installation of solar panels (not shown) and/or for

the construction of the solar farm **104** may be gathered and stored. As depicted in FIG. **1**, the materials **116** may include beams **120**. However, the materials **116** may correspond to and/or include, but are not limited to, beams, torsion bars, rods, columns, channels, fastening systems, solar panels, electrical/electronic devices, wires, cablings, clamps, and the like materials, all or most of which may be needed for the construction and/or maintenance of the solar farm **104**. Other apparatuses and/or systems may be present at the storage area **112**, as well, as shall be understood from the description set out further below. Out of the materials **116** listed above, aspects related to the beams **120** are discussed throughout the present disclosure and are depicted in the figures (not all beams **120** are marked in FIGS. **1**, **8**, and **9** for clarity). Discussions corresponding to the beams **120** may be suitably applied to one or more of the other materials **116**, e.g., similar materials, such as rods, tubes, columns, as may be available at the storage area **112**, at any given point.

[0020] As shown in FIG. **1**, in some embodiments, the construction area **108** may adjoin the storage area **112**. The construction area **108** may receive a supply of the materials **116** from the storage area **112** for the installation of the solar panels and/or for the construction of the solar farm **104** at the construction area **108**. The sites of the construction area **108** at which the solar panels may be installed may be referred to as ‘solar panel sites **124**’. Solar panel sites **124** may correspondingly define predetermined locations **132** at which the materials **116** (e.g., the beams **120**) may be respectively delivered and/or deposited, such that by use of the materials **116**, the solar panels may be correspondingly installed at the solar panel sites **124**. Although not limited, the solar panel sites **124** (and thus the solar panels at those sites) in the construction area **108** may be laid out according to a grid pattern, i.e., in rows and/or columns. Various other layout patterns for the solar panel sites **124** and thus the solar panels, such as a rotational pattern, a polygonal pattern, an oblong pattern, a linear pattern, an irregular pattern, and other patterns, may be contemplated.

[0021] An aspect of the present disclosure discusses, an autonomous or semi-autonomous system and manner of delivery of the beams **120** to the predetermined locations **132** (e.g., one beam **120** per one predetermined location **132**). In this regard, and as an example view provided in FIG. **1**, beams **120** are shown to be deposited at the predetermined locations **132** in corresponding solar panel sites **124** in the construction area **108**—the deposition being executed by an autonomous or semi-autonomous beams delivery system and/or process. In some embodiments, the beams **120** may be deposited in the construction area **108** in rows that may comply with the example grid pattern of the solar panel sites **124**, described above. For example, the beams **120** in the predetermined locations **132** of the solar panel sites **124** of FIG. **1** (which may correspond to the beams **120** in FIG. **4**) may be collectively referred to as a first beam row **128**, and, understandably, the construction area **108** may include one or more other beam rows **128'**, as well.

[0022] The beams **120**, as such, may include, but are not limited to, I-beams and/or H-beams. Although not limited, the beams **120** may include a linear profile and may be formed, at least in part, from a metallic material. Further, the beams **120** may all be congruous and/or identical to each other—for example, the beams **120** may be equal in shape and dimension to each other, although variations in the shape and size of the beams **120** may be contemplated by someone of skill in the art based on the present disclosure. With regard to the transfer of the beams **120** (e.g., the autonomous or semi-autonomous beams delivery system and/or process) to the predetermined locations **132**, an example aspect of the present disclosure describes a system **136** (see FIGS. **11A-C**) for delivering the beams **120** to the predetermined locations **132** of the work site **100**. The system **136** includes a work machine **140**, a sorting assembly **144**, and a control system **340** in operable communication with the work machine **140** and the sorting assembly **144**. As shown in FIGS. **11A-C**, different, non-limiting, embodiments of the system **136** are disclosed, and each of those embodiments are annotated by the same reference numeral, i.e., ‘**136**’, for ease in reference. These embodiments are described later in the present disclosure.

[0023] Referring to FIG. **8**, and also in conjunction with FIGS. **1**, **5**, **6**, and **7**, the sorting assembly **144** is described. The sorting assembly **144** is configured to receive the beams **120** in loose batches

(e.g., from another machine or device) (not shown) and arrange and/or sort the beams **120** in a predefined arrangement **196**. The predefined arrangement **196**, as referred to in the present disclosure, may not necessarily correspond or relate to the number of beams **120**. Rather, the predefined arrangement **196** of the beams **120** may correspond to an arrangement of the beams **120**, at any given point, in which the beams **120** may be positioned one after the other, e.g., in a serial fashion such that a longitudinal side face **148** of one beam **120** (or each beam **120**) (see FIGS. **4** and **9**) may face or be directed towards a longitudinal side face **148** of another beam **120** (or an adjacent beam **120**). The sorting assembly **144** may include a receptacle **152**, a stacking platform **156**, and a conveyor **160**, as shown. Although not limited, the sorting assembly **144**, as a whole, may be provided at the storage area **112** (see FIG. **1**).

[0024] The receptacle **152** may include or correspond to a container, and in some examples, may be in the form of a hopper **164**. The receptacle **152** may include a cavity **168** defining a first opening **172** and a second opening **176**. Beams **120** in loose batches may be introduced (e.g., by any suitable method) into the cavity **168** through the first opening **172** such that the sorting assembly **144** may receive the beams **120**. The receptacle **152** may hold (or store) the beams **120** in loose batches therewithin. Further, the receptacle **152** may release the beams **120** out of the cavity **168** through the second opening **176**. As an example, a release of the beams **120** from the cavity **168** through the second opening **176** may be attained under the action of gravity. Further, as an example, a release of a number of beams **120** from the receptacle **152** through the second opening **176** may be suitably controlled and/or regulated, e.g., by use of a gate or a door (not shown) that may selectively open and close the second opening **176**.

[0025] The conveyor **160** may include an endless belt **180**. The endless belt **180** may operably extend from the receptacle **152** all the way to the stacking platform **156**. The endless belt **180** may receive the beams **120**, e.g., loosely, from the receptacle **152**, may support the beams **120**, and may move, e.g., with respect to a conveyor frame **122**, to convey the beams **120** (from the point of its receipt on the endless belt **180**) to the stacking platform **156**. As an example, a portion of the conveyor **160** and/or the endless belt **180** may be disposed under the second opening **176** of the receptacle **152** such that the endless belt **180** may receive an efflux of the beams **120** (e.g., under the action of gravity) from the receptacle **152**, through the second opening **176**.

[0026] Referring to FIG. **9**, and in some embodiments, the conveyor **160** may include a driver pulley **184** and a driven pulley **188**. The driver pulley **184** and the driven pulley **188** may be spaced apart from each other. The endless belt **180** may be operably wound and supported around each of the driver pulley **184** and the driven pulley **188** and may be sufficiently tensioned therebetween. In so doing, a rotation of the driver pulley **184** (e.g., by using electrical actuation) may cause the endless belt **180** to move or rotate with the rotation of the driver pulley **184**, in turn also causing the driven pulley **188** to rotate (e.g., synchronously) with the rotation of the driver pulley **184**.

[0027] In some embodiments, the endless belt **180** may define a number of grooves **192**. The grooves **192** may be serially arranged one after the other and in regular intervals along a travel path, E, movable by the endless belt **180** from the receptacle **152** to the stacking platform **156**. As an example, the grooves **192** may extend along a width, W, of the endless belt **180** (see FIG. **8**). The grooves **192** may be configured to correspondingly receive and accommodate the beams **120**, at least partially, therewithin, and guide the beams **120** toward the stacking platform **156** and/or the predefined arrangement **196** as the conveyor **160** urges (a section of) the endless belt **180** towards the stacking platform **156**. In one example, one beam **120** may be accommodated in one groove **192**. In some embodiments, the beams **120** may be accommodated within the grooves **192** such that they can, at least partly, span across the width, W, of the endless belt **180**.

[0028] In some embodiments, the conveyor **160** may include a vibratory unit **200** (see FIG. **9**). The vibratory unit **200** may impart vibration to the endless belt **180** to urge the beams **120** to correspondingly settle and be accommodated into the grooves **192**. The vibratory unit **200** may be located within or adjacent to one or more of the driver pulley **184** and/or the driven pulley **188**.

and/or the conveyor frame **122** such that a vibration (or a shaking motion) may be imparted to the endless belt **180**. A working and a functional association of the vibratory unit **200** with the conveyor **160** to impart vibration to the conveyor **160** may be contemplated by someone of skill in the art, and thus will not be discussed in the present disclosure.

[0029] The stacking platform **156** may be used to accumulate and stack the beams **120**. In this regard, the stacking platform **156** may include one or more rails (e.g., see first rail **204** and second rail **208** in FIG. **8**) (also see FIG. **9**). As may be understood, the second rail **208** may be hidden behind the first rail **204** in the illustration of the sorting assembly **144** in FIG. **9**, but the same can be viewable in FIG. **8**). Both the first rail **204** and the second rail **208** may be positioned or disposed relatively close to the endless belt **180**, such that the beams **120** may exit the endless belt **180** and be delivered onto the first rail **204** and the second rail **208** at an end of the travel path, E. In this regard, as the beams **120** may move in a direction away from the receptacle **152** and approach the stacking platform **156**, the beams **120** may disembark or alight from the endless belt **180** and may be pushed onto the first rail **204** and the second rail **208**. In some embodiments, the first rail **204** and the second rail **208** (or portions of the first rail **204** and the second rail **208**) may be positioned lower in elevation to the endless belt **180** such that a debarking of the beams **120** from the endless belt **180** and a subsequent boarding of the beams **120** onto each of the first rail **204** and the second rail **208** may be relatively easily attained.

[0030] Although not limited, both the first rail **204** and the second rail **208** may be positioned at the same elevation with respect to the endless belt **180** or the conveyor **160**. In so doing, the first rail **204** and the second rail **208**, when receiving the beams **120**, may also help stabilize the beams **120** on the stacking platform **156**. In some embodiments, the first rail **204** may define an end **212** disposed towards the stacking platform **156** and relatively away from the endless belt **180**, as shown. A portion **216** of the first rail **204** (and the second rail **208**) defining the end **212** (and the end **212'**), extending beyond the endless belt **180**, may wholly or partly form the stacking platform **156**. In some embodiments, the portion **216** may be tilted at an angle to an extension or a plane of the endless belt **180** such that the end **212** may be positioned lower in elevation to a point at which the beams **120** may exit the conveyor **160** and be pushed onto the first rail **204** (and the second rail **208**). Also, the tilting of the portion **216** may be exaggerated in FIG. **9** for ease in understanding. Description corresponding to the first rail **204** may be suitably comported to the second rail **208**, as well, as applicable.

[0031] An angle at which each of the first rail **204** and the second rail **208** may be tiled or angled (e.g., with respect to the extension of the endless belt **180** or the horizontal) may be the same. Such titled or angular positioning of the portions (e.g., portion **216**) of first rail **204** (and the second rail **208**) may ensure that the beams **120** onboard the first rail **204** and second rail **208** may, freely or without external support, slide and advance under the action of gravity towards the ends **212**, **212'** associated with the first rail **204** and the second rail **208**, and thus may accumulate and stack up one behind the other at said ends **212**, **212'** in the predefined arrangement **196**. In some embodiments, the stacking of the beams **120** at the ends **212**, **212'** may be attained as the beams **120** trailing on the endless belt **180** may push the beams **120** leading on the stacking platform **156**.

[0032] In some embodiments, the stacking platform **156** may include one or more stoppers—e.g., see first stopper **220** and second stopper **224**. The first stopper **220** and second stopper **224** may be respectively positioned at the ends **212**, **212'** of the first rail **204** and the second rail **208** and may be applied to limit the slide or advancement of the beams **120** beyond the ends **212**, **212'** of the first rail **204** and the second rail **208**. In some embodiments, the stacking platform **156** also includes walls **228**, **228**, which may be positioned laterally oppositely to each other, as shown in FIG. **8**, so as to restrict a lateral displacement (or misplacement) of the beams **120** as the beams **120** slide along the portions (e.g., portion **216**) of first rail **204** (and the second rail **208**).

[0033] Referring to FIGS. **2**, **3**, and **4**, the work machine **140** will be now described. The work machine **140** may include a compact track loader **232**, although other machine types may be

contemplated based on the present disclosure. The work machine **140** may be employed to deliver the beams **120** from the sorting assembly **144** (or the storage area **112**) to the construction area **108** or to the predetermined locations **132** of the construction area **108**. As an example, delivering the beams **120** may mean loading the beams **120** onto the work machine **140**, carrying the beams **120** correspondingly up to the predetermined locations **132**, and unloading/depositing the beams **120** correspondingly at the predetermined locations **132**, such that at least one beam **120** is made available at one predetermined location **132**. More particularly, aspects of the present disclosure may correspond to the segment of the construction process in which one beam **120** is delivered per predetermined location **132**. The work machine **140** may correspond to various other work machines, such as, a compact wheel loader, a multi-terrain loader, and/or the like machine, that may perform the function of the work machine **140**, as is described in the present disclosure. Therefore, references to the compact track loader **232** are by example only.

[0034] The work machine **140** may include a chassis or a frame **236**. Further, the work machine **140** may include one or more traction devices **240** supported by the frame **236**. As an example, the traction devices **240** may include crawlers or endless tracks, although the traction devices **240** may also include wheels, provided either alone or in combination with the crawlers. The work machine **140** may define a front end **244** and a back end **248**. The terms ‘front’ and ‘back’, as used herein, are in relation to an example direction of travel of the work machine **140**, and the same is illustratively represented by an arrow, T, in FIG. 2, with said direction of travel being defined from the back end **248** towards the front end **244**.

[0035] The work machine **140**, when moving in the direction, T, defined from the back end **248** towards the front end **244**, may be referred to as a condition in which the work machine **140** moves or travels in a ‘forward direction’. Conversely, the work machine **140**, when moving oppositely to the direction, T, defined from the front end **244** towards the back end **248**, may be referred to as a condition in which the work machine **140** moves or travels in a ‘reverse direction’. A movement of the traction devices **240**, in either the forward direction or the reverse direction, may be powered by a power source **242**, such as an engine and/or an electric motor (see FIG. 10), housed within a power compartment **252** (see FIGS. 2 through 4) of the work machine **140**. Also, the work machine **140** may include an operator cab **256** to house various controls of the power source **242** and/or an implement **260** of the work machine **140**. Apart from the power source **242**, the work machine **140** may also include a number of sub-systems **246** (see FIG. 10) that may assist with machine movement and/or a machine navigation at the work site **100**, such as a braking system, a steering system, a safety system (e.g., including an obstacle detection system), etc.

[0036] The implement **260** may be movable with respect to the frame **236**. In some embodiments, the implement **260** corresponds to or includes a forklift device **264**, which may be generally applicable to carry apparatuses, such as pallets, e.g., inside a shopfloor. The forklift device **264** may include one or more forks, e.g., a first fork **268** and a second fork **272**, as shown. Also, the forklift device **264** (e.g., the first fork **268** and the second fork **272**) may be raisable and lowerable with respect to the frame **236**, for example, by way of an activation of an actuation system **238** (e.g., a hydraulic actuation system) (see FIG. 10) of the work machine **140**. According to an aspect of the present disclosure, the implement **260** may be applied when moving the beams **120** to the predetermined locations **132** of the construction area **108** of the work site **100**, as will be understood from the description further below.

[0037] The work machine **140** may include one or more haul elements **276** mountable onto the implement **260** (see FIGS. 3 and 4) (not all haul elements **276** may be marked in the figures for clarity). In other words, the haul elements **276** may be mounted onto the first fork **268** and the second fork **272** of the implement **260**. As an example illustration in FIGS. 3 and 4, five (5) haul elements **276** are (combinedly) mounted sequentially, one after the other, to the first fork **268** and the second fork **272**. A higher or a lower number of haul elements **276** may be mounted to the first fork **268** and the second fork **272**, and thus the number five (5) is provided only for illustrative

purposes. A first haul element **276'** may be positioned farthest from the frame **236**, while a fifth or last haul element **276''''** may be disposed closest to the frame **236**.

[0038] The haul elements **276** may be used or applied to pick the beams **120** from the stacking platform **156** of the sorting assembly **144** (in the predefined arrangement **196**) and transport the beams **120** to the predetermined locations **132** of the construction area **108** of the work site **100**. As an example, one beam **120** may be grasped by one haul element **276**. Therefore, if five (5) haul elements **276** are mounted to the forklift device **264** (or to the first fork **268** and the second fork **272** of the implement **260**), a maximum of five (5) beams **120** may be grasped by the haul elements **276** and transported from the sorting assembly **144** to corresponding five (5) predetermined locations **132** of the construction area **108** in a single delivery cycle. As an example, see a first beam **120'** being grasped by the first haul element **276'** and a fifth or last beam **120''''** being grasped by the fifth or last haul element **276''''**. Description further below includes some structural discussions corresponding to the first haul element **276'**. Such discussions may be suitably applied to all other haul elements **276**.

[0039] Referring to FIGS. **5**, **6**, and **7**, the first haul element **276'** may include an elongated profile. Further, the first haul element **276'** may define an upper surface **280**, a lower surface **284**, a first side surface **288**, and a second side surface **292**, as shown. The first haul element **276'** may include one or more engaging portions for correspondingly engaging with the forklift device **264**. As an example, the engaging portions include or correspond to a first engaging portion **296** and a second engaging portion **300**. The first engaging portion **296** and the second engaging portion **300** may define corresponding apertures—e.g., a first aperture **304** and a second aperture **308** through which the first fork **268** and the second fork **272** of the implement **260** can correspondingly pass through. In so doing, effectively, the first engaging portion **296** and the second engaging portion **300** of the first haul element **276'** correspondingly engage with the first fork **268** and the second fork **272** of the first haul element **276'**.

[0040] Further, the first haul element **276'** includes one or more separator blocks—e.g., a first separator block **312** and a second separator block **316**. The first separator block **312** and the second separator block **316** may be applied to space out and maintain a clearance (e.g., a constant clearance) between one haul element (e.g., the first haul element **276'**) and an adjacent haul element (e.g., a second haul element **276**) (see FIGS. **3** and **4**). The separator blocks may also help maintain a clearance between the beams **120** held by the haul elements **276** (e.g., adjacent haul elements **276**), at any given point. To this end, the first separator block **312** and the second separator block **316** may be each positioned at the second side surface **292** of the first haul element **276'**. As shown in FIG. **7**, the first separator block **312** may include a stub portion **320** and a tapered portion **324** extending from the stub portion **320**. The tapered portion **324** defines a tip **328** and may define a body with a tapering profile extending till the tip **328**. An extension of the tapered portion **324** towards the tip **328** may be aligned in a direction that extends from the upper surface **280** of the first haul element **276'** towards the lower surface **284** of the first haul element **276'**. At least a part of the tapered portion **324**, along with the tip **328**, may extend outwards and/or beyond a plane of the lower surface **284** of the first haul element **276'**. Similar discussion may be contemplated for the second separator block **316**.

[0041] In some embodiments, as the beams **120** may be made from a metallic material, magnetic action may be used to couple the beams **120** to the haul elements **276**. In this regard, the first haul element **276'** may include an electromagnet **332** (see FIG. **7**) to respectively attract, grasp, and retain at least one beam **120** (e.g., one corresponding beam, see beam **120'**, FIG. **4**), thereto. As an example, the electromagnet **332** may be switchable between various conditions—e.g., a first condition and a second condition. The first condition may correspond to an active condition of the electromagnet **332** and in which the first haul element **276'** may be able to attract and couple a corresponding beam (e.g., beam **120'**) thereto. The second condition may correspond to an inactive condition of the electromagnet **332** and in which the first haul element **276'** may be able to

decouple and/or dislodge the corresponding beam (e.g., beam **120'**) therefrom.

[0042] The work machine **140** may further include a sensor system **336** (FIG. **10**) having a variety of sensors to detect one or more machine parameters. As an example, the sensor system **336** may include one or more of each of a machine speed sensor to sense a ground speed of the work machine **140**; an accelerometer or an inertial measurement unit (IMU) sensor to sense an acceleration/orientation of the work machine **140** and/or the implement **260**; location sensors (e.g., a Global Positioning System (GPS), a Global Navigation Satellite System (GNSS), or any other location tracking system) to detect a location of the work machine **140**; an obstacle sensor to detect an obstacle in a moving path of the work machine **140**; a set of imaging or visual sensors, such as 3d (dimensional) sensors, LIDAR (Light Detection and Ranging) sensors, image/video capturing devices (such as cameras), and the like sensors, to detect the surroundings of the work machine **140**; and one or more pressure sensors to detect a load on the implement **260**. Such sensors may be disposed on various, appropriate parts of the work machine **140** to sense one or more of the above noted parameters associated with the work machine **140**. The sensors are well known in the art and hence, not described in greater detail for the sake of brevity of the present disclosure.

[0043] Referring to FIG. **10**, the system **136** may include the control system **340** for attaining an autonomous or a semi-autonomous functioning of the work machine **140**, enabling the work machine **140** to transfer the beams **120** to the predetermined locations **132** of the construction area **108** of the work site **100**. To this end, the control system **340** may correspond to a main or a single controller, which may be operably coupled (e.g., through a wired connection or wirelessly) to various systems or sub-systems (e.g., sub-systems **246**) of the work machine **140** and to various units or sub-units (e.g., the vibratory unit **200**) of the sorting assembly **144**, described above, such that it can control the working of one or more of the work machine **140** and the sorting assembly **144**. For example, systems or sub-systems of the work machine **140** may include, but are not limited to, the actuation system **238**, the power source **242**, and other sub-systems **246**, as shown in FIG. **10**.

[0044] The control system **340** may include one or more processing units configured to process data received from the systems or sub-systems associated with the work machine **140** and/or from the units or sub-units associated with the sorting assembly **144**. The control system **340** may convert said data into formats readable by one or more other systems or sub-systems associated with the work machine **140** and/or units or sub-units associated with the sorting assembly **144** such that those systems and/or units can take relevant/corrective action in pursuance to the detection of any work machine condition.

[0045] The control system **340** may also be in operable communication (e.g., through a wired connection or wirelessly) with an input device, such as a joystick, touchscreen, associated with the work machine **140** (e.g., those that may be provided within the operator cab **256** of the work machine **140** or those that may be provided outside or away from the work machine **140** from where an instruction for executing various work machine procedures or sorting assembly procedures may be received. As an example, the instruction may be initiated by an operator stationed within the operator cab **256** or stationed remotely to the work machine **140**. As an example, the instruction may correspond to the transfer of the beams **120** from the storage area **112** or from the sorting assembly **144** to the predetermined locations **132**. To effectuate such transfer of the beams **120**, the control system **340** may also be in operative communication with various other systems and/or devices of the work machine **140**, such as the power source **242** of the work machine **140** and the sub-systems **246** of the work machine **140** so as to suitably control those systems to maneuver the work machine **140** when transporting the beams **120**.

[0046] Further, the control system **340** may be in communication (e.g., through a wired connection or wirelessly) with a memory **348**. As some non-limiting examples, the memory **348** may be located within one of the control system **340**, the work machine **140**, or the sorting assembly **144**, or may be located remote to one or more of these. As some further non-limiting examples, the

memory **348** may include multiple individual data storage devices, e.g., three (3) data storage devices, with such data storage devices being correspondingly located within the control system **340**, the work machine **140**, and the sorting assembly **144**. As an example, the memory **348** may store a set of instructions that the control system **340** may retrieve in response to the receipt of the instructions. The control system **340** may be configured to execute the set of instructions retrieved from the memory **348** to perform various steps associated with a particular procedure—e.g., the steps associated with the transfer of the beams **120** from the storage area **112** (or from the sorting assembly **144**) to the predetermined locations **132** of the construction area **108** of the work site **100**. [0047] The control system **340** may also be in operative communication (e.g., through a wired connection or wirelessly) with the electromagnet **332** (e.g., all electromagnets **332** associated with the haul elements **276**) so as to switch (e.g., selectively switch) the electromagnet **332** (or all electromagnets **332**) between the first condition and the second condition, as needed. The memory **348** may also store a map (e.g., a virtual map) of the work site **100** based on which the control system **340** may detect a real-time location of the work machine **140** on the work site **100** and by way of which the control system **340** may also suitably guide the work machine **140** to various locations, e.g., the predetermined locations **132**, of the work site **100**.

[0048] According to example stages of operation of the control system **340**, in response to the instructions, and based on the execution of the set of instructions retrieved from the memory **348**, the control system **340** may be configured to issue a command to move the work machine **140** for an operative engagement with the sorting assembly **144**. To this end, the control system **340** may generate a retrieval command to alter the implement **260** (e.g., by manipulating the actuation system **238** associated with the forklift device **264** to either lower or raise the first fork **268** and the second fork **272**) and operate the power source **242** to suitably move the work machine **140** towards the stacking platform **156** of the sorting assembly **144** so as to be in relative proximity to the predefined arrangement **196**.

[0049] Based on the retrieval command, the control system **340** may switch the haul elements **276** to the first condition (or energize the corresponding electromagnets **332** associated with the haul elements **276** to activate the magnetism of the corresponding electromagnets **332**). In the first condition, the haul elements **276** may be applied to retrieve the beams **120** (e.g., by magnetic force) from the sorting assembly **144** in the predefined arrangement **196**. As an example, a number of the beams **120** retrieved from the sorting assembly **144** may be same as the number of haul elements **276** that may have been switched to the first condition, since those that have been switched to the first condition may be in the appropriate grasping state, e.g., in the magnetically active state to retrieve the beams **120** with magnetic force.

[0050] As the haul elements **276** may grasp the beams **120** in the predefined arrangement **196** from the stacking platform **156** of the sorting assembly **144**, the control system **340** may further provide a notification to haul the retrieved beams **120** to the predetermined locations **132** of the construction area **108** of the work site **100**. As an example, the control system **340** may ensure that at least one beam **120** is hauled to each predetermined location **132** such that at least one beam **120** is deposited at each predetermined location **132**. For this purpose, the control system **340** may also generate release commands correspondingly at the predetermined locations **132** such that at least one haul element **276** switches from the first condition to the second condition (e.g., by de-energizing the corresponding electromagnet(s) **332**) to release at least one beam **120** correspondingly at the predetermined locations **132**.

[0051] In some embodiments, the beams **120** may be released correspondingly at the predetermined locations **132** in preset orientations. More particularly, the beams **120** retrieved by the haul elements **276** and released at the predetermined location **132** may be released on to a ground surface of the construction area **108** of the work site **100** in a manner that they are each deposited on the ground surface in the same orientation.

[0052] In some embodiments, the control system **340** may also control a working of the sorting

assembly **144**. For example, the control system **340** may also be configured to supply an activation signal to the conveyor **160** for conveyor activation (e.g., by electrical actuation) and conveyance of the beams **120** from the receptacle **152** to the stacking platform **156**, in response to the instructions. The control system **340** may be further configured to actuate the vibratory unit **200** in response to the conveyor activation. In addition, as the control system **340** may be communicably coupled (e.g., through a wired connection or wirelessly) to the electromagnets **332**, the control system **340** may also be configured to selectively energize and de-energize the electromagnets **332** to change or toggle the haul elements **276** mounted onto the implement **260** between the first condition and the second condition.

[0053] In some embodiments, the control system **340**, in response to the notification, may be configured to determine a predefined movement sequence for moving the work machine **140** from a first predetermined location **132'** at the construction area **108** to a last predetermined location **132''''** at the construction area **108** (see FIG. 1). In this regard, the control system **340** may be configured to move the work machine **140** according to the predefined movement sequence from the first predetermined location **132'** up to the last predetermined location **132''''** along a machine travel path—e.g., in the reverse direction of the work machine **140**. The predefined movement sequence (and many such sequences, as the need may be to cover a relatively large number of predetermined locations **132** at the construction area **108** of the work site **100**) may be pre-stored in a map (that corresponds to the work site **100**) which the control system **340** may selectively source or retrieve from the memory **348** to move the work machine **140** according to the predefined movement sequence(s).

[0054] In some embodiments, the control system **340** may also determine multiple routes for the work machine **140** at the work site **100**. Such determination may be enabled by way of the map, e.g., between a real time location of the work machine **140** (e.g., from the storage area **112**) all the way to the first predetermined location **132'**, so as to guide the work machine **140** to the first predetermined location **132'** such that once the work machine **140** reaches the first predetermined location **132'**, the work machine **140** may move from the first predetermined location **132'** up to the last predetermined location **132''''** along the machine travel path depositing a beam **120** at each predetermined location **132** along the machine travel path.

[0055] It will be further appreciated that the control system **340** may be configured to generate the release commands to release the beams **120** correspondingly at the predetermined locations **132** according to a release pattern. The release pattern may facilitate a progressive release of the beams **120** from the first beam **120'** to the fifth or last beam **120''''** (see FIGS. 3 and 4) such the first beam **120'** may be released at the first predetermined location **132'** and the fifth or last beam **120''''** may be released at the fifth or last predetermined location **132''''**. Notably, the first beam **120'** may be disposed farthest from the frame **236** of the work machine **140** and the last beam **120''''** may be disposed closest to the frame **236** of the work machine **140**.

[0056] Referring to FIGS. 11A-C, and in some embodiments, the control system **340** may correspond to a machine controller associated with the work machine **140** and may be located, at least in part, within the work machine **140** (as shown in FIG. 11A) and may be part of an electronic control module (ECM) associated with the work machine **140**. In such as case, the machine controller may be in operative communication with a sorting assembly controller of the sorting assembly **144** so to control both the sorting assembly **144** and the work machine **140** in the manner as has been discussed in the present disclosure. In alternative examples, or conversely, the control system **340** may correspond to a sorting assembly controller associated with the sorting assembly **144** (as shown in FIG. 11B), and which may be in operative communication with the machine controller of the work machine **140** so to control both the sorting assembly **144** and the work machine **140** in the manner as has been discussed in the present disclosure.

[0057] Further, alternatively, the control system **340** may correspond to an independent, remote controller stationed altogether outside and/or remote to each of the work machine **140** and to the

sorting assembly **144** (as shown in FIG. **11C**), but which may be in operative communication (e.g., through a wired connection or wirelessly) with the machine controller associated with the work machine **140** and the sorting assembly controller associated with the sorting assembly **144**, such that a working of both the work machine **140** and the sorting assembly **144** may be controlled by the remote controller. Effectively, it may be noted that controller functionality, as described above, may be performed one or more controllers and in case of multiple controllers, said functionality may be suitably shared between the multiple controllers in concerted and/or a collaborative manner to deliver the beams **120** appropriately to the predetermined locations **132** at the construction area **108** of the work site **100**. Such collaboration, and variations thereof between multiple controllers, may be contemplated by someone of skill in the art based on the present disclosure and is thus not discussed.

[0058] The control system **340** may be a microprocessor-based device, and/or may be envisioned as an application-specific integrated circuit, or other logic devices, which provide controller functionality, and such devices being known to those with ordinary skill in the art. In one example, it is possible for the control system **340** to include or be representative of one or many control systems having separate or integrally configured processing units to process a variety of data (or input or commands) and/or to share one or more functionality with the one or more of the other control systems, as has been noted above. In some embodiments, a transmission of data between the control system **340** and various other controllers and/or the sensors, etc., may be facilitated wirelessly or through a wired connection, or other through other methods which are now known or in the future developed.

[0059] Processing units of the control system **340**, to convert and/or process various input, command, signals, and/or the like, may include, but are not limited to, an X86 processor, a Reduced Instruction Set Computing (RISC) processor, an Application Specific Integrated Circuit (ASIC) processor, a Complex Instruction Set Computing (CISC) processor, an Advanced RISC Machine (ARM) processor, or any other processor.

[0060] Examples of the memory **348** may include a hard disk drive (HDD), and a secure digital (SD) card. Further, the memory **348** may include non-volatile/volatile memory units such as a random-access memory (RAM)/a read only memory (ROM), which may include associated input and output buses. The memory **348** may be configured to store various other instruction sets for various other functions of the work machine **140** and/or the system **136**, along with the set of instructions, discussed above.

INDUSTRIAL APPLICABILITY

[0061] Referring to FIG. **12**, an example method for delivering the beams **120** to the predetermined locations **132** of the work site **100** is discussed. The method is discussed by way of a flowchart **1200**, in conjunction with FIGS. **1** through **11A-C**. The method starts at stage **1202**.

[0062] At stage **1202**, as an example segment of a construction process of the solar farm **104**, an operator may provide the instruction, e.g., manually via an input device, to transfer the beams **120** (e.g., correspondingly) to the predetermined locations **132**. The control system **340** may receive and process the instruction. It will be appreciated that prior to the instruction, the haul elements **276** may be mounted onto the first fork **268** and the second fork **272**, e.g., by having the first fork **268** and the second fork **272** correspondingly pass through the first aperture **304** and the second aperture **308** of the first engaging portion **296** and the second engaging portion **300** such that the first fork **268** and the second fork **272** may engage with the first engaging portion **296** and the second engaging portion **300**. The method moves to stage **1204**.

[0063] At stage **1204**, in response to the instruction, the control system **340** may issue a command to move the work machine **140** for operative engagement with the sorting assembly **144**. Based on the command, the control system **340** may actuate the power source **242** and control a movement of the work machine **140** such that the work machine **140** can reach up to the sorting assembly **144** and operatively engage with the sorting assembly **144**. It may be noted that controlling the

movement of the work machine **140**, at any given point, may mean and/or include one or more of: controlling the power source **242** of the work machine **140** and/or controlling the sub-systems **246** (e.g., the braking system, the steering system, the safety system) of the work machine **140**.

[0064] For operative engagement of the work machine **140** with the sorting assembly **144**, the visual sensors, such as image capturing devices, or cameras mounted on the work machine **140** may capture a video feed of the work machine **140** with respect to the sorting assembly **144** and transmit the video feed to the control system **340**. The control system **340** may use one or more image analyzers (not shown) to detect the edges and/or contours of the work machine **140** and the sorting assembly **144** enabling the control system **340** to determine when an orientation and/or a proximity of the work machine **140** with respect to the sorting assembly **144** is properly attained such that the work machine **140** can operatively engage with the sorting assembly **144**.

[0065] Further, as part of stage **1204**, the control system **340** may supply an activation signal to the conveyor **160** for conveyor activation and conveyance of the beams **120** from the receptacle **152** to the stacking platform **156** in response to the instruction. In some embodiments, the control system **340** may perform both, e.g., supply the aforesaid activation signal and issue the command to move the work machine **140** for operative engagement with the sorting assembly **144**, simultaneously. In so doing, at least a first batch of the beams **120** may be stacked (or may be already in process to be stacked) in the stacking platform **156** as the work machine **140** arrives at the sorting assembly **144** for the operative engagement. Further, the control system **340** may also actuate the vibratory unit **200** in response to the conveyor activation such that vibration can be imparted to the endless belt **180** to urge the beams **120** to be correspondingly enter and be accommodated into the grooves **192**.

[0066] In some embodiments, the control system **340** may determine, e.g., through one or more weight sensors (not shown) of the sorting assembly **144**, a weight of the beams **120** stacked at the stacking platform **156** in the predefined arrangement **196**. In such a case, if the weight is in excess of a weight threshold, the control system **340** may determine a condition in which the stacking platform **156** holds sufficient number of beams **120** and may accordingly deactivate conveyor movement and thus the conveyance of the beams **120** towards the stacking platform **156**. The control system **340** may then re-activate the conveyor **160** once the weight recedes below the weight threshold. The method moves to stage **1206**.

[0067] Next, at stage **1206**, and as part of operatively engaging the work machine **140** with the sorting assembly **144**, the control system **340** may generate a retrieval command to alter the implement **260** and switch the haul elements **276** to the first condition to retrieve the beams **120** from the sorting assembly **144** in the predefined arrangement **196**. For this purpose, the control system **340** may actuate the actuation system **238** of the work machine **140** to appropriately raise (and/or lower) the first fork **268** and the second fork **272** to bring the first fork **268** and the second fork **272** up to a level such that the haul elements **276** are at an elevation with respect to the beams **120** stacked in the predefined arrangement **196** in the stacking platform **156**. Thereafter, the control system **340** may control the movement of the work machine **140**, e.g., to advance or retract or steer the work machine **140** appropriately such that the haul elements **276** may be correspondingly suspended above the beams **120** stacked in the predefined arrangement **196** in the stacking platform **156**.

[0068] Thereafter, the control system **340** may lower the haul elements **276** (e.g., by accessing the actuation system **238** associated with the forklift device **264**) such that the haul elements **276** can come closer to the beams **120** to grasp the beams **120**, e.g., one haul element **276** may grasp one beam **120** by magnetic force—see FIG. 4. When lowering the haul elements **276** to retrieve and grasp the beams **120** from the stacking platform **156** (or the sorting assembly **144** in the predefined arrangement **196**) by the haul elements **276**, tips **328** associated with the first separator block **312** and the second separator block **316** may correspondingly allow the first separator block **312** and the second separator block **316** to enter into corresponding spaces defined consecutively between the beams **120** sorted in the predefined arrangement **196** in the stacking platform **156**, causing said

beams **120** to align respectively with the haul elements **276** and facilitating the haul elements **276** to respectively grasp and retain the beams **120** thereto—e.g., the beams **120** may snap correspondingly onto the haul elements **276** owing to the magnetic attraction of the corresponding electromagnets **332** in the haul elements **276**.

[0069] In some embodiments, the implement **260** may include one or more weight sensors, as also described for the stacking platform **156**, that the control system **340** can access to determine whether the beams **120** (e.g., a minimum number of beams **120**) are indeed grasped by the haul elements **276** or not. The method moves to stage **1208**.

[0070] At stage **1208**, the control system **340** may provide a notification to haul the beams **120**, grasped by the haul elements **276** in the predefined arrangement **196**, to the predetermined locations **132**. In some embodiments, pursuant to the notification, the control system **340** may source the map of the work site **100** from the memory **348** and detect a real time location of the work machine **140** (e.g., based on data from the locations sensors) and associate a point on the map corresponding to the real time location of the work machine **140**. The control system **340** may also detect a predetermined location on the map (e.g., which may correspond to the first predetermined location **132'** or any other predetermined location **132** defined along the predefined movement sequence) to which the work machine **140** is to travel up to for depositing (one of) the beams **120**. The control system **340** may further determine a route (e.g., a virtual route on the map) from said point to said predetermined location on the map. Once the route is determined, the control system **340** may control the movement of the work machine **140** at the work site **100** in order to comply with and follow along said route such that the work machine **140** may be moved to the predetermined location **132**. The method moves to stage **1210**.

[0071] At stage **1210**, as the work machine **140** may reach the predetermined location **132** (e.g., the first predetermined location **132'**) at the construction area **108**, the control system **340** may compare the real time location of the work machine **140** (i.e., the point on the map) with the predetermined location on the map. Pursuant to a match between the point and the predetermined location on the map, the control system **340** may determine that the work machine **140** has arrived at the first predetermined location **132** (e.g., the first predetermined location **132'**), and further that at said predetermined location **132** at least one beam **120** (e.g., first beam **120'**) is to be released and/or deposited.

[0072] Pursuant to the match between the point and the predetermined location on the map, the control system **340** may generate a release command at the predetermined location **132** (e.g., the first predetermined location **132'**) such that at least one haul element (e.g., the first haul element **276'**) switches from the first condition to the second condition to release at least one beam (e.g., the first beam **120'**) at said predetermined location **132** (e.g., the first predetermined location **132'**). In this regard, the control system **340** may be configured to selectively energize and de-energize the electromagnet **332** to change the corresponding haul element **276** between the first condition and the second condition. As an example, to release the first beam **120'** from the first haul element **276'**, the control system **340** may de-energize the electromagnet **332** associated with the first haul element **276'** to accordingly change the first haul element **276'** from the first condition to the second condition to release the first beam **120'** at the first predetermined location **132**. Similar discussion may be contemplated for each of the other beams **120** corresponding to the remaining haul elements **276** for the remaining predetermined locations **132**.

[0073] Also, the beams **120** may be released correspondingly at the predetermined locations **132** in preset orientations. With regard to releasing the first beam **120'** in the preset orientation (and also depositing the first beam **120'** on the ground surface accordingly), the work machine **140** may use the IMU sensor (which may be a part of the sensor system **336** of the work machine **140**).

According to an example application, the IMU sensor may be positioned towards the front end **244** of the work machine **140**, that may indicate, e.g., on the map, a direction in which the work machine **140** is facing at any given point. Such indication may be provided in conjunction with the

point on the map that indicates the work machine's real time location. Based on such direction of the work machine **140** on the map, the control system **340** may estimate the orientation that the forklift device **264**, the haul elements **276**, and thus first beams **120** are in, at any given point. [0074] For releasing the first beam **120'** in the preset orientation, as soon as the work machine **140** acquires a predefined orientation state with respect to the first predetermined location **132'** suitable for the release of the first beam **120'**, the control system **340** may detect the said predefined orientation state and generate the corresponding release command (e.g., a first release command) for the release of the first beam **120'** at the first predetermined location **132**. Such description may be applicable for every other beam **120** correspondingly grasped by every other haul element **276** and to every other predetermined location **132** at the construction area **108** of the work site **100**. Although the description related to the detection of the orientation state above is described, it will be appreciated that various other orientation and detection methods, now known or in the future developed, can be applied to detect the orientation of the work machine **140** or various parts of the work machine **140**. The method ends at stage **1210**.

[0075] With regard to the control system **340** also determining the predefined movement sequence for moving the work machine **140** from the first predetermined location **132'** to the fifth or last predetermined location **132''''**, in some embodiments, the predefined movement sequence may be prestored in the memory **348**. With reference to the example layout of the predetermined locations **132** in FIG. 1, a predefined movement sequence for delivering the beams **120** at those predetermined locations **132** may include a straight line machine travel path or sequence, with the work machine **140** moving in the reverse direction from the first predetermined location **132'** to the fifth or last predetermined location **132''''** and stopping for beam delivery at every predetermined location **132** along said straight line machine travel path or sequence.

[0076] Therefore, once the predefined movement sequence is determined, the control system **340** may control the movement of the work machine **140** to move the work machine **140** according to the predefined movement sequence from the first predetermined location **132'** up to the fifth or last predetermined location **132''''** along the machine travel path, e.g., the straight line machine travel path or sequence, in the reverse direction. Further, in process of doing so, the control system **340** may also be configured to generate the release commands to release the beams **120** correspondingly at the predetermined locations **132** (e.g., a first release command to release the first beam **120'** at the first predetermined location **132'**, a second release command to release the second beam **120''** at the second predetermined location **132''**, and so on) according to a release pattern.

[0077] The release pattern may facilitate a progressive release of the beams **120** from the first beam **120'** to the fifth or last beam **120''''**. For this purpose, the control system **340** may sequentially switch the haul elements **276** from the first haul element **276'** to the fifth or last haul element **276''''** correspondingly from the first condition to the second condition starting from the first predetermined location **132'** to the fifth or last predetermined location **132''''**. In effect, the first haul element **276'** is switched to the second condition from the first condition at the first predetermined location **132'**, the second haul element **276''** is switched to the second condition from the first condition at the second predetermined location **132''**, and so on, and such may continue until the fifth or last haul element **276''''** is switched to the second condition from the first condition at the fifth or last predetermined location **132''''**.

[0078] Further, as the work machine **140** may be needed to deliver the beams **120** at multiple other predetermined locations, similar to the predetermined locations **132**, the memory may store (or be prestored with) multiple other predefined movement sequences for the work machine **140** that the control system can access and execute (i.e., the control system **340** may access and control a movement of the work machine **140** accordingly), such that beams **120** at those other predetermined locations can be delivered as well.

[0079] In some embodiments, the work machine **140** may travel anywhere across the work site **100**, as has been described above, while avoiding obstacles along the machine's moving path. In

order to do so, the control system **340** may also (periodically or regularly) obtain data from the sensor system **336**, e.g., the obstacle sensor, such that a presence of an object around or ahead of the work machine **140** may be detected. Upon detecting the presence of the object, the control system **340** may reduce the speed of and/or may altogether stop the work machine **140** in its moving path.

[0080] In some embodiments, when the object is determined to be a person(s) and/or a different machine(s), the control system **340** may stop the work machine **140** for a first predefined period for the person(s) and/or the different machine(s) to move out of the moving path. When the person(s) and/or the different machine(s) is detected to be in the moving path even after a lapse of the predefined period, the control system **340** may determine an alternate moving path for the work machine **140**. When the object is determined to be an inanimate object, the control system **340** may move the work machine **140** around the inanimate object and return to its moving path. If the control system **340** determines a machine or an environmental restriction or obstruction to move the work machine **140** around the inanimate object, the control system **340** may determine an alternate moving path for the work machine **140** to control a movement of the work machine along the alternate moving path.

[0081] An automated manner of delivery of the beams **120** at the predetermined locations **132** of the construction area **108** of the work site **100**, as described above, eases out the overall effort to deliver (and deposit) beams **120** at the predetermined locations **132**. Further, such automated delivery of the beams **120** reduces the time needed to construct the solar farm **104**. Additionally, the system **136** improves safety of work site personnel, enhances work site planning and productivity, and ensures effective repeatability of construction related operations at the work site **100**.

[0082] Unless explicitly excluded, the use of the singular to describe a component, structure, or operation does not exclude the use of plural such components, structures, or operations or their equivalents. The use of the terms “a” and “an” and “the” and “at least one” or the term “one or more,” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The use of the term “at least one” followed by a list of one or more items (for example, “at least one of A and B” or one or more of A and B”) is to be construed to mean one item selected from the listed items (A or B) or any combination of two or more of the listed items (A and B; A, A and B; A, B and B), unless otherwise indicated herein or clearly contradicted by context. Similarly, as used herein, the word “or” refers to any possible permutation of a set of items. For example, the phrase “A, B, or C” refers to at least one of A, B, C, or any combination thereof, such as any of: A; B; C; A and B; A and C; B and C; A, B, and C; or multiple of any item such as A and A; B, B, and C; A, A, B, C, and C; etc.

[0083] It will be apparent to those skilled in the art that various modifications and variations can be made to the method and/or system of the present disclosure without departing from the scope of the disclosure. Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the method and/or system disclosed herein. It is intended that the specification and examples be considered as examples only, with a true scope of the disclosure being indicated by the following claims and their equivalent.

Claims

1. A system for delivering one or more beams to one or more predetermined locations of a work site, the system comprising: a sorting assembly configured to receive the one or more beams in loose batches and arrange the one or more beams in a predefined arrangement; a work machine including an implement having one or more haul elements mounted thereto; and a control system configured to: receive an instruction to transfer the one or more beams to the one or more predetermined locations; issue a command to move the work machine for operative engagement

with the sorting assembly in response to the instruction; generate a retrieval command to alter the implement and switch the one or more haul elements to a first condition to retrieve the one or more beams from the sorting assembly in the predefined arrangement; provide a notification to haul the one or more beams in the predefined arrangement to the one or more predetermined locations; and generate one or more release commands correspondingly at the one or more predetermined locations such that at least one haul element of the one or more haul elements switches from the first condition to a second condition to release at least one beam of the one or more beams correspondingly at the one or more predetermined locations in a preset orientation.

2. The system of claim 1, wherein the sorting assembly includes: a receptacle to hold the one or more beams in loose batches; a stacking platform to accumulate and stack the one or more beams; and a conveyor including an endless belt operably extending from the receptacle to the stacking platform, wherein the control system is configured to supply an activation signal to the conveyor for conveyor activation and conveyance of the one or more beams from the receptacle to the stacking platform in response to the instruction.

3. The system of claim 2, wherein the endless belt defines grooves serially arranged in regular intervals along a travel path of the endless belt from the receptacle to the stacking platform, wherein the grooves are configured to correspondingly accommodate the one or more beams, at least partially, therewithin, and guide the one or more beams toward the predefined arrangement.

4. The system of claim 3, wherein the conveyor includes a vibratory unit to impart vibration to the endless belt to urge the one or more beams to be correspondingly accommodated into the grooves, wherein the control system is configured to actuate the vibratory unit in response to the conveyor activation.

5. The system of claim 1, wherein the implement includes a forklift device including one or more forks raisable and lowerable with respect to a frame of the work machine, each haul element of the one or more haul elements including one or more engaging portions for correspondingly engaging with the one or more forks.

6. The system of claim 1, wherein each haul element of the one or more haul elements includes one or more separator blocks to maintain a clearance between one haul element of the one or more haul elements and an adjacent haul element of the one or more haul elements.

7. The system of claim 6, wherein each separator block of the one or more separator blocks includes a tapered portion defining a tip, wherein to retrieve the one or more beams from the sorting assembly in the predefined arrangement by the one or more haul elements, tips associated with the one or more separator blocks correspondingly allow the one or more separator blocks to enter into corresponding spaces defined consecutively between the one or more beams sorted in the predefined arrangement, causing said one or more beams to respectively align with the one or more haul elements and facilitating the one or more haul elements to respectively grasp and retain the one or more beams thereto.

8. The system of claim 1, wherein each haul element of the one or more haul elements includes an electromagnet to respectively attract and retain at least one beam of the one or more beams thereto, wherein the control system is configured to selectively energize and de-energize the electromagnet to change each haul element of the one or more haul elements between the first condition and the second condition.

9. The system of claim 1, wherein, in response to the notification, the control system is configured to: determine a predefined movement sequence for moving the work machine from a first predetermined location of the one or more predetermined locations to a last predetermined location of the one or more predetermined locations; and move the work machine according to the predefined movement sequence from the first predetermined location up to the last predetermined location along a machine travel path.

10. The system of claim 9, wherein: the control system is configured to generate the one or more release commands to release the one or more beams correspondingly at the one or more

predetermined locations according to a release pattern, the release pattern facilitates a progressive release of the one or more beams from a first beam of the one or more beams to a last beam of the one or more beams, and the first beam is disposed farthest from a frame of the work machine and the last beam being disposed closest to the frame of the work machine.

11. A method for delivering one or more beams to one or more predetermined locations of a work site, the method comprising: receiving, by a control system, an instruction to transfer the one or more beams to the one or more predetermined locations; issuing, by the control system, a command to move a work machine for operative engagement with a sorting assembly in response to the instruction, the sorting assembly configured to receive the one or more beams in loose batches and arrange the one or more beams in a predefined arrangement, the work machine including an implement having one or more haul elements mounted thereto; generating, by the control system, a retrieval command to alter the implement and switch the one or more haul elements to a first condition to retrieve the one or more beams from the sorting assembly in the predefined arrangement; providing, by the control system, a notification to haul the one or more beams in the predefined arrangement to the one or more predetermined locations; and generating, by the control system, one or more release commands correspondingly at the one or more predetermined locations such that at least one haul element of the one or more haul elements switches from the first condition to a second condition to release at least one beam of the one or more beams correspondingly at the one or more predetermined locations in a preset orientation.

12. The method of claim 11, wherein the sorting assembly includes: a receptacle to hold the one or more beams in loose batches; a stacking platform to accumulate and stack the one or more beams; and a conveyor including an endless belt operably extending from the receptacle to the stacking platform, the method further including: supplying, by the control system, an activation signal to the conveyor for conveyor activation and conveyance of the one or more beams from the receptacle to the stacking platform in response to the instruction.

13. The method of claim 12, wherein the endless belt defines grooves serially arranged in regular intervals along a travel path of the endless belt from the receptacle to the stacking platform, wherein the grooves are configured to correspondingly accommodate the one or more beams, at least partially, therewithin, and guide the one or more beams toward the predefined arrangement.

14. The method of claim 13, wherein the conveyor includes a vibratory unit to impart vibration to the endless belt to urge the one or more beams to be correspondingly accommodated into the grooves, the method further including: actuating, by the control system, the vibratory unit in response to the conveyor activation.

15. The method of claim 11, wherein the implement includes a forklift device including one or more forks raisable and lowerable with respect to a frame of the work machine, each haul element of the one or more haul elements including one or more engaging portions for correspondingly engaging with the one or more forks.

16. The method of claim 11, wherein each haul element of the one or more haul elements includes one or more separator blocks to maintain a clearance between one haul element of the one or more haul elements and an adjacent haul element of the one or more haul elements.

17. The method of claim 16, wherein: each separator block of the one or more separator blocks includes a tapered portion defining a tip, wherein to retrieve the one or more beams from the sorting assembly in the predefined arrangement by the one or more haul elements, tips associated with the one or more separator blocks correspondingly allow the one or more separator blocks to enter into corresponding spaces defined consecutively between the one or more beams sorted in the predefined arrangement, causing said one or more beams to respectively align with the one or more haul elements and facilitating the one or more haul elements to respectively grasp and retain the one or more beams thereto.

18. The method of claim 11, wherein each haul element of the one or more haul elements includes an electromagnet to respectively attract and retain at least one beam of the one or more beams

thereto, the method further including: selectively energizing and de-energizing, by the control system, the electromagnet to change each haul element of the one or more haul elements between the first condition and the second condition.

19. The method of claim 11 further including: determining, by the control system, a predefined movement sequence for moving the work machine from a first predetermined location of the one or more predetermined locations to a last predetermined location of the one or more predetermined locations; and moving, by the control system, the work machine according to the predefined movement sequence from the first predetermined location up to the last predetermined location along a machine travel path.

20. The method of claim 19, wherein: the control system is configured to generate the one or more release commands to release the one or more beams correspondingly at the one or more predetermined locations according to a release pattern, the release pattern facilitates a progressive release of the one or more beams from a first beam of the one or more beams to a last beam of the one or more beams, and the first beam is disposed farthest from a frame of the work machine and the last beam being disposed closest to the frame of the work machine.
